Vessel-Guided Curriculum Learning with Uncertainty Calibration for Retinal Disease Classification on RFMiD

K Dhiraj
School of Computer Science and Engineering (SCOPE)
Vellore Institute of Technology
Chennai, India
dhiraj.k2023@vitstudent.ac.in

M Vaishnavi
School of Computer Science and Engineering (SCOPE)
Vellore Institute of Technology
Chennai, India
vaishnavi.m2023@vitstudent.ac.in

Apurva Sahay
School of Computer Science and Engineering (SCOPE)
Vellore Institute of Technology
Chennai, India
apurva.sahay2023@vitstudent.ac.in

Dr. Modigari Narendra
School of Computer Science and Engineering (SCOPE)
Vellore Institute of Technology
Chennai, India
modigari.narendra@vit.ac.in

***Abstract*—Retinal fundus screening could widen access to eye care. Real datasets, though, hold many diseases and only a few images of the rarer ones, so a model that looks accurate overall often misses the very conditions that matter most. This paper asks whether three additions to a standard convolutional classifier help in that long-tailed, multi-label setting. The first is anatomical guidance from a frozen retinal-vessel segmentation model. The second is a soft curriculum that weights training samples by class and sample difficulty. The third is post-hoc calibration with Monte Carlo (MC) dropout, which reports how far each prediction can be trusted. We test them on the RFMiD dataset, using five-fold multilabel-stratified cross-validation, paired-bootstrap confidence intervals and a held-out test set.**

The strongest model was a well-trained DenseNet-121 with none of these additions (out-of-fold mAP 0.634; test-ensemble mAP 0.622). Vessel guidance produced no detectable change, and the curriculum reduced accuracy. Diagnostic checks showed that the network never used the vessel prior as intended. Calibration, by contrast, pushed expected calibration error (ECE) below 0.006 for every model tested, and deferring the most uncertain images to a human reader improved performance on the rest. Rare diseases stayed the main weakness. Several were never detected at the operating thresholds, and their test results rest on only a handful of positives. The paper's main contributions are a reproducible evaluation protocol for long-tailed fundus classification, a calibrated and uncertainty-aware baseline, and documented negative results on vessel guidance and curriculum learning.

***Index Terms*—retinal fundus classification, long-tailed learning, vessel guidance, curriculum learning, uncertainty calibration, RFMiD**

I. Introduction

Multi-disease fundus screening is only useful if it stays reliable on rare diseases and says how far each prediction can be trusted. Colour fundus photography is cheap and widely available, but trained graders are scarce in underserved regions, which is what motivates automated multi-label classification [1]. The Retinal Fundus Multi-disease Image Dataset (RFMiD) [15] supports the task with 3,200 images, each carrying a normal/abnormal label and 45 disease labels. It is also one of the hardest public benchmarks. Labels overlap, and the most common retained class outnumbers the rarest by $138:1$. A classifier can therefore post good aggregate scores while missing the very diseases that matter clinically.

Two ideas look attractive here. The first concerns the retinal vessels, which show up in every image and are altered by many diseases, so a vessel segmentation map works as a disease-agnostic anatomical prior [2], [5]. The second is a curriculum that postpones emphasis on hard and rare examples, which can help long-tailed training [17], [18]. Post-hoc calibration and MC-dropout uncertainty, meanwhile, can tell a reader which predictions to distrust [19], [20]. The literature seldom combines these ideas, and seldom tests them rigorously (Section II).

We built a pipeline that brings them all together. Because tail classes have only a handful of test positives, we put as much effort into measurement as into modelling. The pipeline has four parts: a frozen HRM-UNet vessel prior, a DenseNet-121 classifier with learnable vessel gates, a two-level soft curriculum with deferred re-weighting (DRW), and a post-hoc calibration chain with MC-dropout triage.

Contributions

1) **A reproducible evaluation protocol** for long-tailed multi-label fundus classification. It specifies a 25-label task and uses five-fold multilabel-stratified cross-validation on 2,560 pooled images, with calibration

and thresholds fitted on out-of-fold predictions, a held-out 640-image test set, and paired-bootstrap confidence intervals. The median number of tail-class positives rises from 4 (test split) to 20 (pooled OOF).

2) **A vessel-guided DenseNet-121** with zero-initialised, learnable residual gates at three scales. The classifier starts out as the pretrained backbone and is free to suppress the prior.
3) **A two-level soft curriculum** whose pacing weight never excludes a sample, along with an analysis of how it failed.
4) **A calibrated, uncertainty-aware multi-label baseline**, combining temperature and class-wise scaling with MC-dropout deferral, with ECE computed on equal-mass bins.
5) **Controlled negative results.** Vessel guidance gave no detectable gain, and the curriculum hurt. Higher input resolution and a fundus-pretrained ViT-L/16 didn't help in our settings, and the best model turned out to be a plain DenseNet-121.

Section II reviews the literature and Section III describes the method. Section IV covers the experimental setup, Section V the results, and Section VI discusses them. Section VII concludes.

II. Literature Review

This section reviews 20 recent studies (2024–2026) on deep-learning-based retinal fundus analysis. They cover disease classification, class imbalance, vessel-guided learning, curriculum learning, calibration, and evaluation practice. Table I lists each study's task, method, reported result, and limitations. Results are quoted as published; because datasets, class definitions, splits, and metrics differ, direct comparison across studies is not possible.

A. Deep Learning for Fundus Disease Classification

Transfer-learned convolutional networks dominate the field. A recent survey of lightweight models [1] places classic CNNs at 90–96% accuracy and MobileNet- and EfficientNet-class networks at 85–94%. It also singles out RFMiD as the most challenging benchmark due to its 46 labels, label co-occurrence, and severe imbalance. Still, reported accuracy reflects task formulation more than algorithmic superiority: balanced, curated single-label datasets yield 93.9–99.8% accuracy [15]–[17]. In contrast, RFMiD-derived tasks fare worse, reaching 75.4% on eight classes [8] and 88.5% on ten classes [11]. A six-label RFMiD formulation achieves 97.3% accuracy, but at a recall of only 0.43 [18].

Backbone rankings do not generalize across studies. DenseNet-121 was the top performer among four networks in a controlled RFMiD comparison [11] and beat EfficientNet in a matched ablation (94% vs. 92.9%) [16]. Yet, it was the weakest component of a four-network ensemble on RFID50K (93.19%) [15]. ResNet-50 reached 97.27% accuracy in [18] but failed to converge in [20]. Ensembles [8], [15] and two-stage architectures [6], [17] yield marginal improvements over strong single models (+0.05 percentage points in [15]) and leave tail classes underperforming; CLARE-Net [8] exhibits a sensitivity of just 0.37 on the diabetic neuropathy (DN) class. Vision Transformers have also been explored via Swin-T feature extractors [17] and standard ViTs [20]. However, neither work provides a tuned CNN baseline under an equivalent protocol, and the ViT in [20] achieves 93.21% training accuracy while dropping to 78.91% on validation.

B. Class Imbalance and Long-Tailed Labels

Studies tackling class imbalance rely primarily on static remedies. These include validation-weighted ensembles and per-class thresholds [8], focal loss [7], [10], weighted binary cross-entropy [7], targeted augmentation of rare classes [11], and SMOTE or synthetic oversampling to equalize class distribution paired with mixup and label smoothing [15], [17]. This latter group artificializes the data distribution, effectively removing the long tail rather than resolving it.

The only systematic evaluation of loss functions and re-weighting schemes on RFMiD [18] compared inverse-frequency weighting, weighted cross-entropy, focal loss, and Tversky loss. It revealed that these choices merely shifted the operating point along a fixed precision-recall trade-off at a default 0.5 threshold: moving from recall 0.43 at precision 0.79, to near-perfect recall at precision 0.04, or to perfect precision at near-zero recall. Deferred re-weighting—a cornerstone of general long-tail literature—remains unapplied in retinal imaging studies. Furthermore, rare-class gains are seldom isolated via ablation or validated on sufficient test samples; in [11], several rare classes contain only two or three test images.

C. Vessel-Guided Learning and Segmentation

The retinal vasculature is visible in every fundus image and is pathologically altered by numerous conditions, making it a natural anatomical prior. However, it has been leveraged almost exclusively for single-disease diabetic retinopathy (DR) grading. Bano *et al.* [2] feed vessel and lesion maps from an Attention U-Net into a CNN for five-grade DR (90% accuracy). Kadhim *et al.* [5] concatenate classical vessel features with CLAHE features to achieve 96.7%, compared to 72–88% for single-feature baselines. A hybrid ResNet-UNet fuses segmentation and classification via dual-pathway attention, reaching 85.17% accuracy [6]. In all cases, the vessel signal acts as a static input for a DR-only task. Multi-label studies omit vessel information entirely, and no existing framework permits the classifier to dynamically modulate its dependence on vessel features, even though many conditions (e.g., drusen) are non-vascular.

Segmentation literature provides potential priors. HRM-UNet [4] reports a DRIVE F_1 score of 0.7988 and sensitivity of 0.7707, while ELA-U-Net [14] reaches an F_1 of 0.8276 with only 0.34M parameters. Other approaches explore semi-dilation supervision for thin vessels [12], semi-supervised learning with adaptive uncertainty thresholds [3], and explainability-driven segmentation [9]. Preprocessing also

plays a key role: green-channel extraction with CLAHE increased image entropy from 2.82 to 3.65 and edge-based contrast from 7.61 to 24.09, enabling a Residual U-Net to reach a Dice score of 0.819 on a pooled four-dataset patch set [19]. However, patch-level metrics are not directly comparable to full-image evaluations.

D. Curriculum Learning, Uncertainty, and Explainability

None of the 20 reviewed studies implements a sample- or class-level curriculum schedule. The closest approximations are pixel-level semi-dilation supervision [12] and layer-wise unfreezing schedules [16], [18], both of which structure model parameters rather than training samples.

Model calibration is similarly under-explored. CLARE-Net [8] is the sole classification study to report Expected Calibration Error (ECE $= 0.0649$), but it lacks an uncalibrated baseline. Reference [15] reports AUC exclusively, whereas [18] applies a global 0.5 threshold across all labels. Standard post-hoc calibration techniques—such as temperature scaling, class-wise calibration, and Monte Carlo dropout—have not been evaluated on long-tailed retinal datasets. Explainability is widely featured but remains qualitative: Grad-CAM variants appear in [7], [10], [11], [15], [16], [20] as visual overlays on selected images, without quantitative validation against anatomical ground truth such as the vessel tree.

E. Evaluation Practice and Research Gaps

Methodological flaws in evaluation practice frequently obscure true performance. Test splits are often small (165 images in [11], 422 in [16], and 643 in [17]), leaving rare classes with as few as two or three test instances. No reviewed study reports multi-seed evaluations or confidence intervals, and only one study tests cross-dataset generalization (94% falling to 88.7%) [16]. Headline metrics routinely mask tail-class failure: 88.48% overall accuracy against a macro-F_1 of 0.697 [11], 82.26% accuracy against a macro-F_1 of ~ 0.60 [10], and 93.93% accuracy against a macro-average accuracy of 89.7% [17]. In [20], a training accuracy of 93.0% contrasts sharply with a validation accuracy of 78.91%. Finally, RFMiD is inconsistently configured across the literature—redefined as 8, 10, or 46 classes, a six-label set [18], or pooled into a four-class set [16]—rendering direct published comparisons unreliable.

In summary, current literature establishes competitive baselines (e.g., DenseNet-121, EfficientNet, standard data augmentation, and weighted loss functions) and demonstrates the utility of vessel features for single-task DR grading. However, existing work fails to integrate four key capabilities: label=(v)

1) Vessel-guided fusion that scales to multi-disease classification;
2) Sample difficulty pacing that systematically prioritizes rare tail classes;
3) Class-wise calibrated uncertainty estimation; and
4) Tail-aware, statistically rigorous evaluation protocols.

These unaddressed gaps directly motivate the methodology introduced in this paper.

III. Methodology

The fundus image is converted into two spatially aligned views, one for the classifier and one for the frozen vessel-segmenting model. The vessel map is inserted into intermediate DenseNet-121 features through learnable gates, the logits are calibrated post hoc, and Monte Carlo (MC) dropout supplies an uncertainty score. The curriculum acts only on the training loss and plays no part at inference (Fig. 1).

A. Data and Labels

RFMiD has 1,920 training, 640 validation, and 640 test images, each with a Disease_Risk label and 45 disease labels (46 in total). Labels with fewer than ten positives in the original training split are dropped after loading, leaving 25 evaluable labels (Fig. 2, Table II). Retained labels are grouped by training prevalence into 8 head, 9 mid, and 8 tail classes. The max-min ratio imbalance is $138:1$.

B. Preprocessing

Each image is decoded once and cropped to the retinal field of view of intensity threshold 10 as roughly 30% of raw pixels are background. The classifier view applies Contrast Limited Adaptive Histogram Equalization (CLAHE; clip limit 2.0, 8×8 tiles) to the LAB luminance channel only. The segmenter view is plain grayscale at 512×512 with no CLAHE, matching the way HRM-UNet was trained on DRIVE. Classifier images are cached at 448 px and vessel maps at 256×256.

C. Frozen Vessel Prior

The prior is an HRM-UNet [4], a 3.71M-parameter model pretrained on the DRIVE dataset [21]. It has a patch embedding, four Visual State Space stages built from Mamba-style selective-scan (SS2D) blocks with Haar-wavelet downsampling, residual-path skips, and a patch-expanding decoder. The network stays frozen, so its maps are computed once and cached.

The selective scan is sequential, which means cost grows with sequence length: 4,416 steps for a 64×64 patch, against 282,624 for a full 512×512 image. For that reason, inference is tiled exactly as it was in training. We use 64×64 patches at stride 32, average the overlaps with unfold/fold, and then downsample the result to 256×256. Hyper-parameters are recovered from the checkpoint tensor shapes, and weights load strictly. If anything does not match, an error is raised, so no layer is ever left randomly initialized.

D. Vessel-Gated DenseNet-121

The classifier used is an ImageNet-initialized DenseNet-121 [22]. After each of the first three transition layers (128, 256, and 512 channels; 28×28, 14×14, and 7×7 maps at 224 px), a gate computes an attention tensor from the vessel map V, average-pooled to the feature size, and applies it residually:

$$A = \sigma\left(\mathrm{MLP}_{1\times1}\left(\mathrm{AvgPool}(V)\right)\right), \quad F_{\text{out}} = F \cdot (1 + \gamma A). \tag{1}$$

TABLE I
SUMMARY OF THE 20 REVIEWED STUDIES

Study	Task and data	Method	Key result
[1] Al-johani et al., 2026	Survey of lightweight retinal classifiers, 2020–2025	PRISMA-style review	CNNs 90–96%, hybrids 92–97%, lightweight 85–94% accuracy
[2] Bano et al., 2025	5-class DR grading; LVPEI, DRIVE	Attention U-Net vessel/lesion maps, then CNN	Accuracy 90%, F1 88%
[3] Hou et al., 2024	Vessel segmentation; STARE	Semi-supervised, adaptive uncertainty	Dice 83.66%
[4] Huang and Yu, 2024	Vessel segmentation; DRIVE	HRM-UNet (VM-UNet, Haar downsampling)	Accuracy 0.9529, sensitivity 0.7707, F1 0.7988
[5] Kadhim et al., 2024	5-class DR; 36:1 imbalance	CLAHE + classical vessel features, CNN	96.7% vs 72–88% baselines
[6] Meenakshi et al., 2025	DR detection; APTOS 2019	Hybrid ResNet–UNet, dual-pathway attention	Accuracy 85.17%, AUC 0.946
[7] Anusha et al., 2026	Multi-label; RFMiD, 46 labels (37 evaluated)	Two-tier EfficientNetB0 + CBAM	Tier-2 per-class accuracy 0.97
[8] Perumalsamy et al., 2026	8-class single-label; RFMiD	EfficientNet-B3 + V2-S adaptive ensemble	Accuracy 75.38%, macro-F1 0.7428, ECE 0.0649
[9] Sadhukhan et al., 2025	Explainable vessel segmentation	[to complete]	[to complete]
[10] Sekhar et al., 2026	DR detection and 5-grade severity; APTOS 2019	DenseNet-121, focal loss, Grad-CAM	Binary accuracy 89.09%; severity 82.26%
[11] Venkatappa et al., 2026	10-class single-label; RFMiD 1.0 + 2.0	ResNet-34, DenseNet-121, VGG-19, Inception-v4	DenseNet-121 accuracy 88.48%, macro-F1 0.697
[12] Wang C. et al., 2024	Vessel segmentation; DRIVE	Semi-dilation supervision, boundary reconstruction	Sensitivity 84.65%, accuracy 96.9%, AUC 98.84%
[13] Wang J. et al., 2024	4-class OCT; 500 training images	Frozen ResNet18, DenseNet121, InceptionV3 features	[to complete]
[14] Zhou and Xu, 2025	Vessel segmentation; DRIVE	ELA-U-Net (0.34 M parameters)	Accuracy 0.9644, sensitivity 0.8139, F1 0.8276
[15] Rahman and Rahman, 2026	7-class single-label; RFID50K	Four-CNN soft-voting ensemble, SMOTE	Accuracy and F1 99.82%
[16] Rokon et al., 2026	4-class; ODIR-5K + RFMiD 2.0	DenseNet121, selective fine-tuning	Accuracy 94%, macro-F1 0.94
[17] James et al., 2026	5-class fundus classification	Swin-T embeddings + MLP	Test accuracy 93.93%
[18] Acharya et al., 2025	Multi-label; RFMiD, 6 label columns	ResNet-50, weighted losses	Accuracy 97.27%, recall 0.43
[19] Sandhya and Nalini, 2026	Vessel segmentation; DRIVE, STARE, CHASE_DB1, HRF	U-Net variants, green-channel CLAHE	Residual U-Net Dice 0.819
[20] Benjamin et al., 2026	10-class eye disease; Mendeley + ODIR-5K	ViT vs CNN baselines	Validation accuracy 78.91% (training 93.21%)

The 1×1 MLP maps one channel to a hidden width of 16 and then to the channel count of F. The scalar γ and the final bias are initialized to zero, the training starts from the pretrained backbone model, and vessel guidance is learned as a departure from it; a negative γ lets the network suppress the prior, which matters because roughly half of RFMiD conditions are non-vascular. The head is global average pooling, dropout ($p = 0.3$), and one logit per retained class, with independent sigmoids because findings co-occur.

E. Long-Tail Loss and Two-Level Soft Curriculum

The loss is per-label binary cross-entropy. Positive weights are capped inverse-frequency terms (cap of 20), computed only from each fold's training labels, and re-weighting is deferred: weights sit at one for the first half of training, then ramp up [23]. Because the loss is kept per sample, the curriculum gets its difficulty signal at no extra cost.

Sample difficulty blends four percentile-ranked signals; ranking by percentile stops any one raw scale from dominating. The first is class rarity, taken from the sample's rarest positive label (0.15). The second is label cardinality (0.15). The third, and the heaviest, is an exponential moving average of per-sample loss (0.55; EMA 0.7). The last is vessel misalignment, defined as one minus the per-sample correlation between gate attention and the vessel map (0.15). Pacing works as a soft weight, not a hard pool:

$$d = 0.30\, d_{\text{class}} + 0.70\, d_{\text{sample}},$$
$$w = \max\left(0.1,\, \sigma\left(\frac{p - \text{pct}(d)}{0.12}\right)\right). \tag{2}$$

Here p is the pace. It rises from 0.60 to 1.0 by 75% of training, after a 25% warm-up, and $\text{pct}(d)$ is the sample's difficulty percentile within its class. Samples easier than the pace get a weight near 1, while harder ones decay toward a floor of 0.1. None is ever excluded. Every epoch therefore visits every sample once, so the curriculum and non-curriculum arms get identical numbers of gradient steps.

Samples with no disease label (about 21% of RFMiD) receive median rarity. Scoring them zero was tried and failed: it filled the warm-up pool with empty images, and the model collapsed to all-negative output.

F. Calibration and Uncertainty

Calibration is applied after training and it preserves rank within each class. Temperature scaling comes first and learns a single scalar T. Class-wise scaling in a diagonal-Dirichlet form then learns a scale and an offset for every class, since miscalibration differs sharply between common and rare classes:

$$z_{\text{cal},c} = a_c \left(\frac{z_c}{T} \right) + b_c. \tag{3}$$

Both calibrators are fitted with L-BFGS on out-of-fold (OOF) logits, and they are cross-fitted (i.e., the ones applied to fold k are fitted on the other four folds).

For uncertainty, dropout in the head stays active at inference while batch normalization remains in evaluation mode. Twenty stochastic passes give a per-class mean and standard deviation, and the mean standard deviation across classes is what ranks images for deferral [24]. Per-class decision thresholds maximize F_1 on a 49-point grid over the OOF predictions. Test data never enters that search.

IV. Experimental Setup

A. Evaluation Protocol

An earlier single-split protocol could not separate real effects from noise. Most tail classes had only four or five test positives, and the observed vessel gain ($+0.012$ mAP) was smaller than the seed-to-seed spread. Consequently, the evaluation protocol was redesigned.

1) Pooling and Hold-Out: Training and validation images are pooled, yielding 2,560 images in total. The official 640-image test split is touched only for final scoring.

2) Stratified Five-Fold CV: The folds use iterative stratification for multilabel data [25]. Samples are visited rarest-label first, and each is assigned to the fold with the highest demand for that label. Every class ends up spread within ± 1 positive across folds (enforced programmatically), with each fold containing 512 images.

3) Out-of-Fold (OOF) Prediction: Each fold model trains on four-fifths of the pool, selects its best epoch on the held-out fifth, and then predicts both that fifth and the test split. Pooled OOF predictions cover all 2,560 images. This yields roughly four times as many positives per tail class as the test split, with a median of 20 against 4.

4) Paired Bootstrap: OOF images are resampled 300 times, and the mAP of two recipes is recomputed on each shared resample. One recipe counts as better or worse only if the 95% confidence interval (CI) of the difference excludes zero.

5) Test Scores: Test numbers come from the five-fold ensemble (mean logits), with calibrators and thresholds fitted on the pooled OOF predictions. The best epoch is chosen on each held-out fold, so OOF scores run mildly optimistic. Test-split numbers were never used for model selection and are treated as unbiased; both are reported.

B. Recipes

Each training run is defined by a single recipe record that is hashed so interrupted runs resume without repeating work (Table III). Recipe C adds curriculum training on top of Recipe B.

TABLE II
Training Recipes

Recipe	Configuration	Folds
A_v7_regime	DenseNet-121, BCE, 30 ep., light regularisation	5
A_baseline	DenseNet-121, BCE, 20 ep., strong regularisation	5
B_vessel	A_baseline + 3 vessel gates	5
C_vessel_curric	B_vessel + soft curriculum	5
B_vessel_320/384px	B_vessel at higher resolution	5
A/B_retfound	RETFound ViT-L/16 [29], without / with gates	1
E0–E5	Vessel + ASL + mixup + SWA, one change each	3

The E-models isolate the effect of individual changes that were previously bundled: E1 re-tunes ASL ($\gamma_- : 4 \to 3$, clip: $0.05 \to 0.08$) [26], E2 introduces a 15% BCE warm-up, E3 relaxes mixup ($\alpha : 0.2 \to 0.15$, $p : 0.5 \to 0.4$) [27], E4 adds rarity-weighted oversampling, and E5 applies gentler learning rates with three warm-up epochs. E0 incorporates Stochastic Weight Averaging (SWA) [28] over the final 25% of training.

C. Training and Implementation

DenseNet-121 is trained with AdamW [30] using a weight decay of 0.05 (biases and normalization parameters are exempt). The backbone learning rate is 10^{-4}, while the gates and classification head use 10^{-3}. Layer-wise learning rate decay is set to 0.8, following a cosine decay schedule after a single warm-up epoch. Batch size is 32, gradient clipping is capped at 5.0, and training operates under mixed precision with the loss maintained in fp32. The best epoch on the held-out fold is retained. Early stopping is disabled because validation mAP exhibits noise (± 0.015), with optimal epochs fluctuating between 15 and 27 across folds.

Data augmentation is executed on the GPU. Spatial transformations, including rotation ($\pm 15^\circ$), zoom, translation, and horizontal flips are applied jointly to the image and its corresponding vessel map. Photometric transformations affect the RGB image only: channel gain ($\pm 6\%$), saturation ($\pm 15\%$), mild blur, and Gaussian noise ($\sigma \leq 0.02$). Cutout and random erasing are avoided because either could eliminate localized lesions defining rare classes. Vertical flips are omitted as fundus image orientation is anatomically consistent.

All experiments were executed in a single environment on an NVIDIA Tesla T4 GPU (15.6 GB) under PyTorch 2.11.0. Per-fold execution times were 3.9 min for the baseline (A) and 4.8 min for vessel gating (B), amounting to a 25–30% per-epoch overhead. The A v7 regime required 5.5 min per fold. RETFound was the most computationally intensive at 18.9 min per fold, approximately eight times slower per epoch than DenseNet-121.

D. Metrics

Accuracy is excluded as a primary quality metric because a naive all-negative classifier achieves over 95% accuracy across all labels. Instead, we report macro mean Average

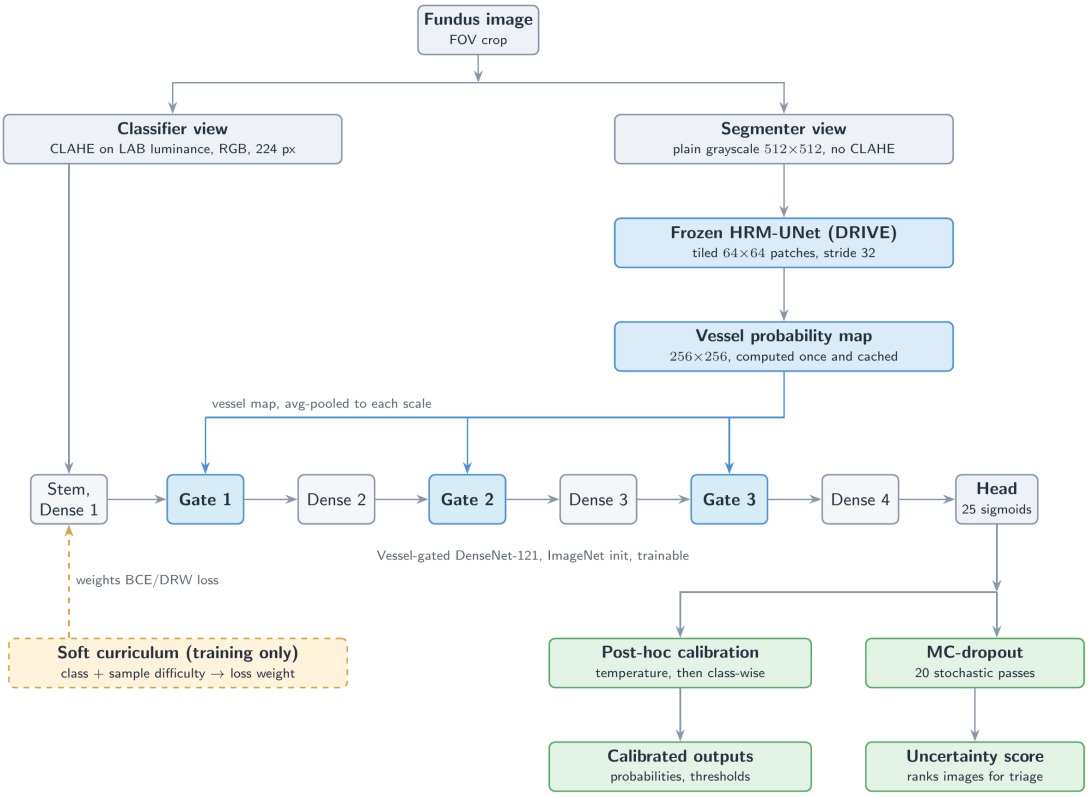


Fig. 1. Proposed Architecture

Precision (mAP) computed over classes with at least one positive sample, which is well-suited for rare-positive detection where ROC-AUC can be overly flattering. Macro AUC is included for completeness and comparability. Additional reported metrics include: head, mid, and tail mAP; macro-F_1 at OOF-fitted decision thresholds; sensitivity and specificity for the binary screening label (normal vs. abnormal) and diabetic retinopathy; and calibration performance, measured via Expected Calibration Error (ECE) over 15 equal-mass bins alongside the Brier score.

Equal-width bins degenerate in this setting because the vast majority of (sample, class) pairs are true negatives predicted near zero. Furthermore, pooled ECE fails to reflect miscalibration on rare classes, prompting us to report ECE grouped by prevalence tier.

Grad-CAM [31] activations are computed from the final normalization layer. Spatial agreement is quantified as the Pearson correlation coefficient between each class activation map and the vessel prior.

V. Results

The best model turned out to be the older 30-epoch DenseNet-121. Vessel guidance made no detectable difference, and the curriculum did clearly worse. Out-of-fold (OOF) metrics (n = 2,560) carry the comparison, because each tail class has a median of 20 pooled positives. Test metrics

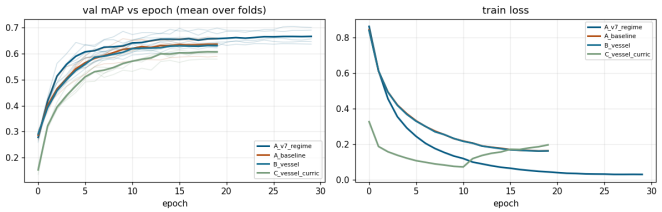


Fig. 2. Validation mAP (mean over folds, faint lines per fold) and training loss for the Stage 1 recipes.

(n = 640, five-model ensemble) play a secondary role: tail classes have only 3-7 positives in the test split.

A. Main comparison

The fold-to-fold standard deviation (SD) of OOF mAP is 0.025, 0.026, 0.018, and 0.019 respectively. Validation mAP rises sharply over roughly five epochs, then levels off and wobbles by ±0.01–0.02 from one epoch to the next (Fig. 3). Training loss under the new regime settles near 0.16 (BCE). Under the older one it dropped to 0.03.

B. Regularization Regime

The 30-epoch, lightly regularized regime outperformed the 20-epoch regime, even though the latter was built to narrow the generalization gap. OOF mAP was 0.634 against 0.607. Compared with the baseline, it came out ahead on 19 of 25

TABLE III
STAGE 1 RESULTS: OUT-OF-FOLD ($n = 2{,}560$) AND TEST-SPLIT ENSEMBLE ($n = 640$).

Recipe	OOF mAP	Head	Mid	Tail	OOF AUC	Test mAP	Test AUC	Test macro-F1
A_v7_regime	0.6340	0.7926	0.6449	0.4631	0.9416	0.6217	0.9541	0.5197
A_baseline	0.6069	0.7731	0.6189	0.4271	0.9408	0.5770	0.9496	0.5185
B_vessel	0.5996	0.7797	0.6056	0.4127	0.9423	0.5766	0.9485	0.5153
C_vessel_curric	0.5757	0.7397	0.5767	0.4108	0.9178	0.5581	0.9221	0.4942

TABLE IV
PAIRED BOOTSTRAP ON OOF MAP (95% CI).

Comparison	Δ OOF mAP [CI]	Verdict	Δ tail mAP
A_v7 vs A_baseline	+0.0271 [+0.0106, +0.0453]	Better	+0.0359 [−0.0088, +0.0820]
B_vessel vs A_baseline	−0.0073 [−0.0201, +0.0055]	No clear diff.	−0.0144 [−0.0527, +0.0241]
C_curric vs A_baseline	−0.0311 [−0.0477, −0.0148]	Worse	−0.0164 [−0.0558, +0.0305]

classes, including 5 of 8 tail classes. One caveat: the two regimes differ in epoch count as well as regularization, with five changes bundled together, so the exact cause can't be isolated.

C. Vessel Guidance

Three vessel gates moved OOF mAP by -0.007 (95% CI $[-0.020, +0.006]$). You can't tell that from zero, and it's smaller than the fold SD of about 0.02. Test-ensemble mAP barely budged (0.5766 against 0.5770). Class by class, vessel guidance raised OOF average precision in 11 of 25 classes, but in only 2 of the 8 tail classes. PT ($+0.079$), ARMD ($+0.026$) and ODP ($+0.022$) gained the most. CRS (-0.061), MS (-0.060) and AH (-0.043) lost the most.

On the test split, DR sensitivity went from 0.742 to 0.831, though specificity slipped (0.971 to 0.954). Macro-F_1 and tail F_1 ended up a little lower (0.515 against 0.519, and 0.337 against 0.366). So the modest, consistent gain we'd seen earlier didn't hold up under cross-validation.

D. Curriculum Learning

Layering the curriculum onto Recipe B (vessel) cut OOF mAP by 0.031 relative to the baseline (95

Tail OOF mAP (0.411) sits close to the baseline's (0.427). On classes with at least five positives, test-ensemble tail mAP (0.513) is higher than the baseline's (0.494). That subset is small, though, and no CI was computed.

Soft pacing did solve the earlier exclusion problem. Every one of the 25 classes received gradient updates from epoch 0, where the earlier hard pool reached only 4 of 46. The vessel-confidence signal, however, was effectively constant (mean $= -0.996$, SD $= 0.002$).

E. Input resolution and a fundus-pretrained backbone

Raising the resolution lifted test-ensemble mAP (0.577 to 0.605) and AUC (0.9485 to 0.9547). The price was about 2.5 times the cost per fold, with no detectable OOF difference to show for it. The threshold value of macro-F_1 and tail-F_1 went the other way and fell as resolution rose.

On fold 0 (512 held-out images), the fundus-pretrained RETFound ViT-L/16 reached a validation mAP of 0.483 without gates and 0.487 with them. DenseNet-121 got 0.641, and Recipe B (vessel) got 0.624. The gaps to Recipe B (-0.141 and -0.137 mAP) have bootstrap CIs that exclude zero ($[-0.188, -0.082]$ and $[-0.191, -0.082]$). Validation mAP was still creeping upward when the 12-epoch budget ran out. That makes this a negative result under the tested settings, not a definitive verdict on the backbone. Gates made no significant difference on RETFound (0.4833 against 0.4869).

F. Individual training changes

Milder mixup and a re-tuned ASL both helped. The ASL warm-up clearly hurt, and two other changes made no detectable difference overall. Rarity oversampling produced the biggest tail gain ($+0.086$ tail mAP, 95% CI $[0.0002, 0.1513]$, right at the edge of significance), yet overall mAP didn't improve and test AUC was the lowest in the group (0.912). That's what you'd expect if the model were simply memorizing the few tail images. Set against plain vessel-guided BCE on the shared folds, nothing in the ASL + mixup + SWA family came out ahead. E0, E2, and E5 were worse on overall mAP, and E1, E3, and E4 couldn't be told apart from it.

G. Calibration

Post-hoc calibration was fitted on OOF predictions and applied cross-fitted. It pushed pooled ECE below 0.006 for every recipe, cutting raw ECE of 0.035–0.075 by roughly an order of magnitude. The ASL and mixup recipes started much worse, with a raw ECE of 0.14–0.21, yet they too ended up at 0.0015–0.0034 in OOF. Because calibration preserves rank within a class, mAP and AUC stay exactly where they were.

The recipe with the lowest raw ECE (A v7 regime) was also the most accurate. The curriculum recipe, by contrast, was the worst calibrated before correction. For Recipe B vessel at 384 px, the five-model ensemble reaches an ECE of 0.0045 and a Brier score of 0.0199 on the test split (Fig. 4).

H. Test-Split Performance and Per-Class Behavior

Per-label accuracy (about 0.97) is inflated by true negatives and should not be read as quality. Macro-F_1 and macro

TABLE V
INPUT-RESOLUTION SWEEP FOR B_VESSEL (FIVE-FOLD).

Input	OOF mAP	Tail mAP	Test mAP (ens)	Test macro-F1	Test tail-F1	Δ OOF mAP vs 224 px [95% CI]
224 px	0.5996	0.4127	0.5766	0.5153	0.3366	reference
320 px	0.6085	0.4232	0.5936	0.5051	0.2701	+0.0089 [−0.0081, +0.0230]
384 px	0.6010	0.4021	0.6045	0.4853	0.2206	+0.0014 [−0.0168, +0.0217]

TABLE VI
ONE-AT-A-TIME ABLATION ON A VESSEL + ASL + MIXUP + SWA BASE (FOLDS 0–2).

Recipe (vs E0)	OOF mAP	Tail mAP	Test mAP (ens)	Δ mAP vs E0 [95% CI]	Verdict
E0 base (vessel + ASL + mixup + SWA)	0.5796	0.3660	0.5530	reference	n/a
E1 ASL re-tuned	0.5982	0.4060	0.5685	+0.0186 [+0.0006, +0.0288]	Better
E2 BCE warm-up for ASL	0.5142	0.3089	0.4727	−0.0654 [−0.0965, −0.0413]	Worse
E3 milder mixup	0.6093	0.4341	0.5755	+0.0297 [+0.0056, +0.0430]	Better
E4 rarity oversampling	0.5905	0.4522	0.5628	+0.0109 [−0.0200, +0.0342]	No clear difference
E5 gentler learning rate	0.5802	0.3795	0.5492	+0.0006 [−0.0169, +0.0143]	No clear difference

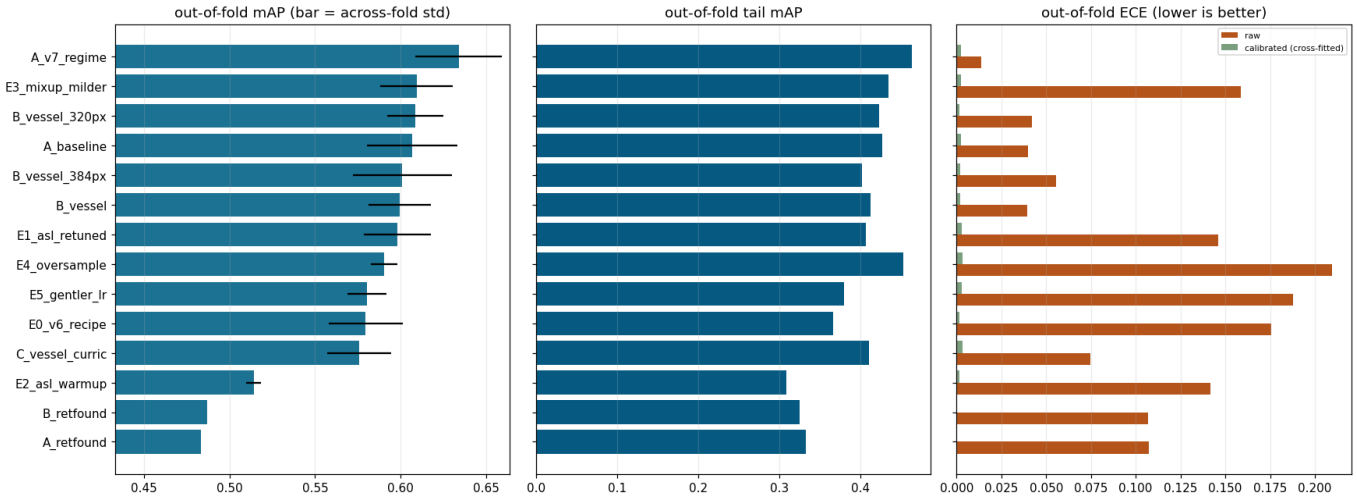


Fig. 3. Out-of-fold mAP (error bar = across-fold SD), tail mAP and ECE before and after cross-fitted calibration for all recipes.

TABLE VII
ECE (EQUAL-MASS BINS) BEFORE AND AFTER POST-HOC CALIBRATION.

Recipe	OOF raw	OOF cal.	Test raw	Test cal.
A_v7_regime	0.0138	0.0026	0.0124	0.0057
A_baseline	0.0398	0.0024	0.0355	0.0042
B_vessel	0.0393	0.0021	0.0349	0.0047
C_vessel_curric	0.0746	0.0033	0.0750	0.0032
B_vessel_320px	0.0422	0.0017	0.0379	0.0034
B_vessel_384px	0.0554	0.0020	0.0513	0.0045

sensitivity, both below 0.52, describe behavior on diseases far better. For the highest-mAP model (Recipe B vessel 384 px; bootstrap 95% CIs over the 640 test images), macro AUC is 0.955 $[0.940, 0.967]$, micro-F_1 is 0.784 $[0.765, 0.802]$, and macro-F_1 is 0.485 $[0.427, 0.529]$. Normal-versus-abnormal screening reached an AUC of 0.977, sensitivity of 0.935, and specificity of 0.858. Diabetic retinopathy reached an AUC of 0.982 and an F_1 of 0.835. The full per-class table is provided in Appendix A.

Performance is strong on the large head classes (DR, MH, Disease Risk: $F_1 \in [0.82, 0.95]$) and falls steeply as class size shrinks. Four classes have zero sensitivity and zero F_1 (RPEC, MS, RT, PT), and three more sit at or below an F_1 of 0.25 (ERM, AION, EDN). Yet their ranking quality is high, with AUCs ranging from 0.92 to 1.00; RT, with five positives, reaches 1.000. In other words, the model ranks rare-disease images well, but thresholds chosen on OOF predictions do not turn that ranking into correct decisions on the test split. The head class DN (46 test positives) is a notable weak point, with sensitivity of 0.33 and an AUC of 0.89.

I. Interpretability and Vessel Agreement

Grad-CAM maps from the fold-0 Recipe B model correlated with the frozen vessel prior at a mean of 0.182 for DR (vascular) and 0.286 for macular hole (non-vascular). The expected pattern, with higher agreement for vascular classes, did not materialize. The learned gate coefficients were negative

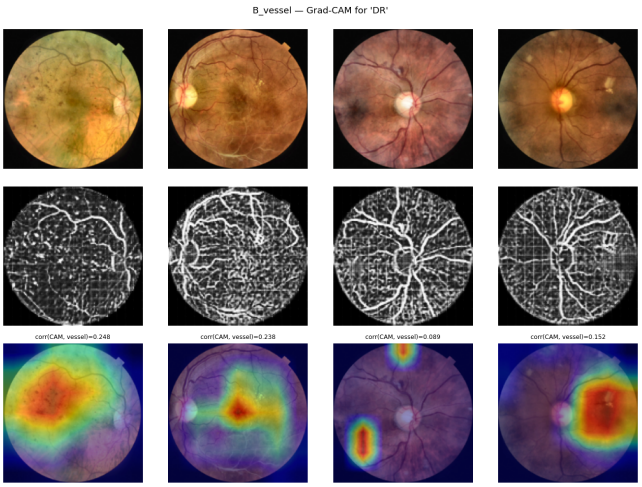


Fig. 4. Grad-CAM for diabetic retinopathy from the fold-0 B_vessel model: image, vessel prior and CAM overlay, with CAM–vessel correlation above each overlay.

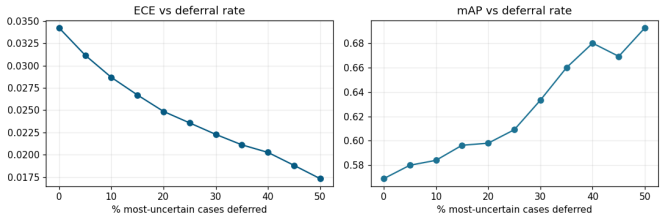


Fig. 5. ECE and mAP on retained test images as the most uncertain cases are deferred (fold-0 B_vessel, 20 MC-dropout passes).

at all three scales: $\gamma = -0.103$ (after transition 1), -0.222 (transition 2), and -0.270 (transition 3). Under this fusion, the network used the prior on average as a suppressive, nuisance signal. These checks rest on a single model and two classes, so they indicate a lack of empirical support rather than a refutation.

J. Uncertainty-Based Deferral

We applied MC-dropout (20 passes) to the fold-0 Recipe B model on the test split and ranked images by mean predictive standard deviation. Deferring the most uncertain images to a human reader raised mAP on the remaining images from 0.569 with no deferral to 0.598 at 20% deferral and 0.693 at 50%. ECE fell from about 0.034 to 0.017 (Fig. 6). Thus, the MC-dropout spread carries meaningful information about image difficulty. Some caution is needed, however: this is evaluated on a single model split with no random-deferral baseline, and mAP over a shrinking set with few tail positives becomes increasingly noisy, so the 50% figure warrants careful interpretation.

K. Comparison with Published Work

Published RFMiD studies report macro-F_1 of 0.70–0.74 and accuracy of 75–88%; in one case ECE is reported as 0.065 [8], [11]. Our macro-F_1 (0.485 for B vessel 384 px, 0.519 for A baseline) and ECE (0.0045) cannot be compared like-for-like. The cited works evaluate 8–10 classes, often considering only single-label accuracy. In contrast, we evaluate 25 multi-label classes, eight of which are rare, using equal-mass ECE bins. The closest direct point of comparison is normal-versus-abnormal screening (AUC 0.977, sensitivity 0.935, specificity 0.858). Our DR AUC (0.982) is comparable in magnitude to published APTOS-2019 benchmarks [6], [10], albeit evaluated on a different dataset.

VI. Discussion

The simplest configuration, a DenseNet-121 trained for 30 epochs with binary cross-entropy, matched or beat every more elaborate variant. Of the nine hypotheses we set out to test, only the two on calibration and uncertainty triage were clearly supported (Table IX).

A. Why the added components did not pay off

Three observations help explain the negative results, though none was tested directly.

1) **Data scale.** Each fold has about 2,000 training images, and each tail class about 20 pooled positives. A component therefore has to clear the fold-to-fold spread (SD about 0.02 mAP) before anyone can detect it. Effects below 0.03 sit at the limit of detection even with 2,560 OOF predictions. That's why cross-validation and the paired bootstrap were essential: without them, we'd have reported noise as progress.
2) **Regularisation.** Training loss under the new regime stayed near 0.16, against 0.03 for the older one, and its validation mAP plateaued lower. Five regularisers combined with a shorter schedule plausibly restricted fitting more than they restricted overfitting.
3) **Fusion design.** Gate attention is computed from the vessel map alone. It adds a learned spatial weighting but carries no image-dependent information. The network chose negative γ, so on average the prior acted as a nuisance signal. A fairer test would fuse the vessel information in an image-dependent way, or apply it only to vascular classes.

The curriculum failure has a specific cause. Its vessel-agreement term was nearly constant (SD 0.002), which made one of the four difficulty signals pure noise. The other three (rarity, cardinality, loss EMA) come close to what DRW and class-balanced weights already encode. A related hypothesis from earlier development remains untested: the curriculum's continuous rarity up-weighting may stack with DRW and produce over-confident predictions on the sparsest classes.

The strongest positive result is calibration. A simple post-hoc scaling fitted on OOF predictions removes most of the miscalibration, even when the raw probabilities were badly distorted (raw ECE 0.14–0.21 for the ASL and mixup recipes). Calibration can therefore be treated separately from the choice of training loss.

TABLE VIII
COMPARISON WITH PUBLISHED WORK

Study	Dataset and task	Acc.	Macro-F1	AUC	Sens. / Recall	ECE
Perumalsamy et al. (2026)	RFMiD, 8-class	0.754	0.743	–	–	0.065
Sekhar et al. (2026)	APTOS, binary DR	0.891	–	–	0.989	–
Sekhar et al. (2026)	APTOS, 5-grade DR	0.823	about 0.60	–	–	–
Venkatappa et al. (2026)	RFMiD, 10-class	0.885	0.697	–	–	–
Anusha et al. (2026)	RFMiD, 37-label multi-label	0.97	–	–	–	–
Bano et al. (2025)	LVPEI, 5-grade DR	0.900	–	–	0.890	–
Hybrid ResNet-UNet	APTOS, DR	0.852	–	0.946	–	–
This work	**RFMiD, 25-label multi-label**	–	**0.485**	**0.955**	**0.935**	**0.002 to 0.006**

TABLE IX
OUTCOME OF EACH HYPOTHESIS

Hypothesis	Evidence	Outcome
New regime narrows generalisation gap	ΔmAP -0.027	No
Vessel guidance improves accuracy	ΔmAP -0.007 (CI spans 0)	No
Vessel prior used more on vascular classes	CAM–vessel corr. DR $0.18 <$ MH 0.29	Improves some vessel-specific classes
Curriculum improves tail performance	ΔmAP -0.031	Improves some tail classes, worse overall
Higher resolution helps small lesions	ΔmAP $\leq$ +0.009 (CI spans 0)	Inconclusive
Fundus-pretrained ViT beats DenseNet-121	-0.14 mAP (fold 0)	No
Post-hoc calibration improves probabilities	ECE $\leq 0.075 \rightarrow <$ 0.006	Yes
MC-dropout is useful for triage	mAP $0.569 \rightarrow 0.693$ at 50% deferral	Yes (preliminary)
Training changes can be separated	Mixup, ASL helped; warm-up hurt	Partly

B. Development history

Earlier iterations of the pipeline, a DenseNet-121 variant run with three seeds, pointed the same way and shaped the final design. One was a rarity-scoring bug. It gave label-free images zero rarity, so they dominated the warm-up pool and the model collapsed. That's why those images now receive median rarity. Another was a mismatch in step budgets, which had left the curriculum arm with fewer gradient steps than the baseline; the soft weighting removes it.

Restricting evaluation to classes with enough training positives raised mAP by about 0.13 across all conditions. That is a measurement artifact, not an improvement, and it's why the filter is stated explicitly.

A hard-exclusion curriculum and the later soft-weighting curriculum both underperformed the baseline on every seed, even though tail mAP beat the baseline in those runs. Temperature scaling alone worsened ECE in every seed-by-condition run. Class-wise scaling, on the other hand, cut ECE three- to sevenfold, hence the final calibrator: temperature first, then class-wise.

These earlier numbers came from a different protocol and aren't directly comparable with Section V. We cite them only for the design lessons.

C. Limitations and threats to validity

- **Vessel guidance and curriculum.** The best recipe (A_v7_regime) uses neither. Vessel guidance was never tested under the better 30-epoch regime, and five regularisation changes were bundled together, so what caused the accuracy loss remains unidentified.
- **Rare-class thresholds.** Four of the 25 classes have zero test sensitivity, and three more have F1 $\leq$ 0.25. Thresholds fitted on OOF predictions don't transfer reliably to rare classes.
- **Few tail positives.** Tail-class test metrics rest on only 3–7 positives, and the 95% CI on macro-F1 is wide (0.427–0.529).
- **Label filtering.** Restricting RFMiD to 25 classes leaves out 21 rare diseases. The mAP values are therefore best compared within this study.
- **Internal validity.** With a single seed per recipe, differences below about 0.02 mAP can't be trusted with much confidence. The E-family used three of the five folds, and the RETFound trial just one. Hyper-parameters weren't tuned per method, which may disadvantage some of them, RETFound included. OOF scores are also mildly optimistic, since the best epoch is chosen on the held-out fold.
- **Construct validity.** Macro metrics are unstable for rare classes, and thresholded F1 depends on thresholds fitted on a different split. One test-mAP value for B_vessel_384px differs slightly between two evaluation cells (0.6029 and 0.6045). We didn't investigate the discrepancy.
- **External validity.** All evaluation is on RFMiD, with a single vessel segmenter trained on DRIVE and applied without adaptation. Nothing was validated externally across cameras or populations, and published work reports 10–20% accuracy loss under domain shift [1]. The interpretability and deferral analyses each use one model.

VII. CONCLUSION AND FUTURE WORK

Calibration and uncertainty-based deferral were the dependable gains in this study. Vessel guidance and curriculum learning did not improve long-tailed multi-label classification on RFMiD. We fused a frozen HRM-UNet vessel prior into DenseNet-121 through zero-initialised gated attention, then added a two-level soft curriculum, class-wise calibration

and MC-dropout. The best model turned out to be a plain DenseNet-121 trained for 30 epochs (OOF mAP 0.634, test-ensemble mAP 0.622, macro AUC 0.954). At 384 px, diabetic retinopathy reached an AUC of 0.982 and an F1 of 0.835. Normal-versus-abnormal screening reached a sensitivity of 0.935, a specificity of 0.858 and an AUC of 0.977.

The central contribution is methodological. The protocol combines five-fold multilabel-stratified cross-validation, OOF-fitted calibration and thresholds, a held-out test split and paired bootstrap intervals. It raised the median number of tail positives from 4 to 20, and it let us state each claim alongside its uncertainty. Reporting the negative results, in turn, gives follow-up work a trustworthy place to start.

Future work, ordered by how directly it follows from the results:

1) **Repair the curriculum.** Replace the constant vessel-confidence signal with a per-sample Grad-CAM-to-vessel correlation, and run a curriculum-without-DRW ablation to test whether stacking rarity weights with re-weighting caused the degradation.
2) **Test whether the prior is used.** Add control runs with shuffled or blank vessel maps, redesign the gates given the negative learned γ (image-dependent fusion, or vascular classes only), and try richer priors (optic disc, macula, lesions) and a segmenter adapted beyond DRIVE.
3) **Strengthen the statistics.** Repeat cross-validation with five to ten seeds, complete the loss and augmentation ablation on all folds, and run the 45-epoch and combined recipes to check interactions.
4) **Address the tail directly.** Recover the 21 excluded classes with few-shot learning, grouped labels or synthetic data, and select thresholds with confidence intervals or cost-sensitive criteria instead of F1 maximisation on a few positives.
5) **Validate beyond RFMiD.** Evaluate the frozen pipeline on multi-centre datasets such as ODIR-5K after mapping categories to the RFMiD label space, and examine robustness across cameras and populations.

APPENDIX A
PER-CLASS TEST RESULTS (B_VESSEL_384PX)

Five-model ensemble on the 640-image test split, with OOF-fitted calibration and per-class F1-optimal thresholds. Tail classes have only 3–7 positives, so individual values are very uncertain.

TABLE X
PER-CLASS TEST METRICS, B_VESSEL_384PX.

Class	Grp	Pos	Thr	Sens	Spec	Prec	F1	AUC
Disease_Risk	head	506	0.50	0.935	0.858	0.961	0.948	0.977
DR	head	124	0.46	0.798	0.973	0.876	0.835	0.982
MH	head	104	0.44	0.875	0.950	0.771	0.820	0.969
ODC	head	91	0.28	0.692	0.918	0.583	0.633	0.909
TSLN	head	53	0.28	0.717	0.951	0.567	0.633	0.967
DN	head	46	0.26	0.326	0.963	0.405	0.361	0.888
MYA	head	32	0.56	0.750	0.995	0.889	0.814	0.995
ARMD	head	31	0.28	0.710	0.977	0.611	0.657	0.972
ODP	mid	24	0.16	0.458	0.976	0.423	0.440	0.948
BRVO	mid	23	0.26	0.739	0.984	0.630	0.680	0.947
ODE	mid	17	0.44	0.706	0.998	0.923	0.800	0.975
LS	mid	15	0.46	0.467	1.000	1.000	0.636	0.937
RS	mid	14	0.20	0.643	0.990	0.600	0.621	0.988
CSR	mid	13	0.32	0.308	0.997	0.667	0.421	0.977
CRS	mid	11	0.30	0.273	0.995	0.500	0.353	0.949
CRVO	mid	9	0.40	0.556	1.000	1.000	0.714	0.987
RPEC	mid	4	0.16	0.000	0.992	0.000	0.000	0.925
MS	tail	7	0.24	0.000	0.995	0.000	0.000	0.941
ERM	tail	5	0.04	0.600	0.975	0.158	0.250	0.922
AH	tail	5	0.46	0.400	0.998	0.667	0.500	0.994
RT	tail	5	0.50	0.000	1.000	0.000	0.000	1.000
AION	tail	4	0.24	0.250	0.998	0.500	0.333	0.937
PT	tail	4	0.36	0.000	0.998	0.000	0.000	0.975
EDN	tail	4	0.06	0.250	0.991	0.143	0.182	0.946
MHL	tail	3	0.08	0.333	1.000	1.000	0.500	0.863

APPENDIX B
PER-CLASS OUT-OF-FOLD AVERAGE PRECISION

AP from OOF predictions over all 2,560 pooled images. Vessel guidance improved 11 of 25 classes and 2 of 8 tail classes against A_baseline; the curriculum improved 4 of 25; the older regime improved 19 of 25.

TABLE XI
PER-CLASS OOF AVERAGE PRECISION.

Class	Grp	Pos	A_base	B_ves	C_cur	A_v7
Disease_Risk	head	2025	0.988	0.989	0.982	0.987
DR	head	508	0.908	0.909	0.862	0.922
MH	head	419	0.866	0.874	0.861	0.875
ODC	head	354	0.722	0.731	0.649	0.757
TSLN	head	251	0.742	0.745	0.718	0.751
DN	head	184	0.411	0.398	0.327	0.420
ARMD	head	138	0.689	0.714	0.672	0.753
MYA	head	135	0.859	0.877	0.848	0.875
BRVO	mid	96	0.777	0.765	0.677	0.808
ODP	mid	91	0.367	0.389	0.318	0.407
ODE	mid	79	0.765	0.740	0.779	0.798
LS	mid	64	0.665	0.681	0.614	0.759
RS	mid	57	0.875	0.860	0.884	0.883
CSR	mid	48	0.633	0.619	0.603	0.603
CRS	mid	43	0.590	0.528	0.482	0.618
CRVO	mid	36	0.799	0.779	0.782	0.794
RPEC	mid	28	0.099	0.090	0.051	0.134
AION	tail	22	0.548	0.571	0.512	0.627
ERM	tail	21	0.103	0.070	0.081	0.062
AH	tail	20	0.735	0.692	0.794	0.809
MS	tail	20	0.188	0.128	0.105	0.127
EDN	tail	20	0.289	0.259	0.232	0.219
RT	tail	20	0.844	0.810	0.837	0.851
MHL	tail	14	0.315	0.298	0.332	0.459
PT	tail	13	0.395	0.474	0.393	0.551

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